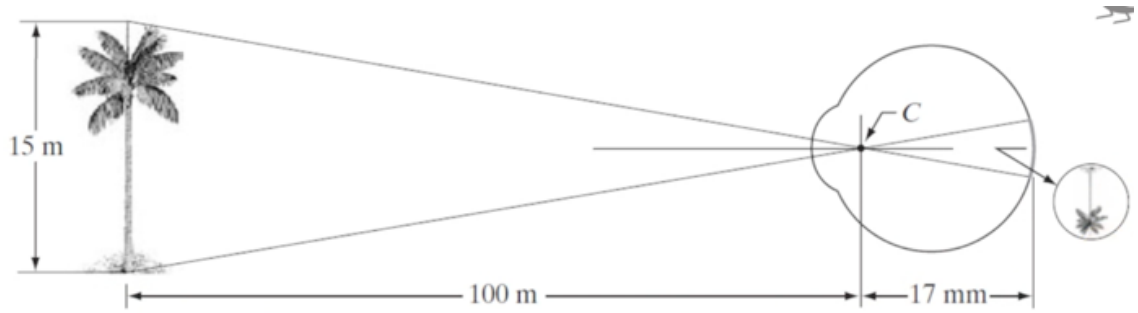
1) explain the image formation in the eye. What are the differences in focusing between the ordinary cameras and the human eyes?

the image formation in the eye : visible range of human eye : 350 nm < λ > 780 nm
for a given λ
f(x,y) denotes images as two-dimensional functions
f intensity value ($f \geq 0$) , $0 < f(x,y) < \infty$
(x,y) spatial coordinates
$f(x,y) = i(x,y)\, r(x,y)$

$0 < i(x,y) < \infty$ is illumination – the amount of source illumination incident on the scene being viewed $0 < r(x,y) < 1$ is reflectance – the amount of illumination reflected by the objects in the scene

Retina contains two types of photoreceptors :
- 100 millions of rods which are distributed over Retina (dark environment)

- 6.5 millions of cones which are mostly packed in Fovea and responsible for color vision .
The number of cones in fovea is about 265K

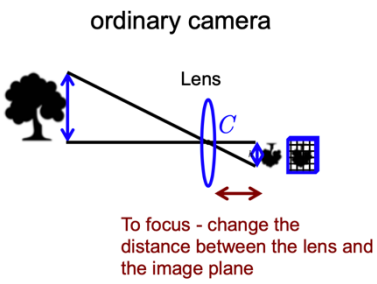


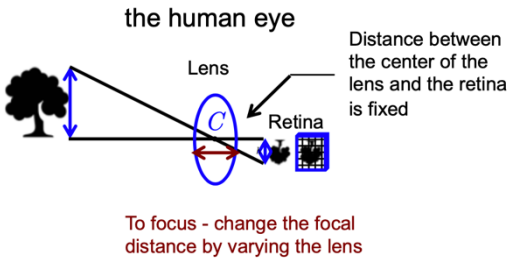
For example suppose that a person is looking at a tree 15 m high at a distance of 100 m. Letting H denote the height of that object in the retinal image, the geometric yields 15/100 = h/17 → h = 2.55 mm

In other words, how human visual system works as below:
1. Light energy is focused by the lens of the eye into sensors and retina.
2. The sensors respond to the light by an electrochemical reaction that sends an electrical signal to the brain (through the optic nerve)
3. The brain uses the signals to create neurological patterns that we perceive as images.

Different focusing between technical camera system and HVS ?
1. First different is the lens. In the HVS lens is flexible but for the most of cameras technique lent is fix, for more expensive cameras we can adapt for focusing but is not flexible like that.

2. The density of photo sensitive element in HVS is not uniform But in the camaras The density of pixels distribution of the senser is always uniform and with the largest concertation in the area of Fovea ,It is very tiny area with size 0.15 mm.





3. Another difference of the HVS and ordinary cameras consists in the way how it perceives the light.
   - ➔ Range of light intensity levels to which HVS can adapt: on the order of 10^{10}.
   - ➔ Intensity as perceived by the HVS(subjective brightness) is a logarithm function of the light intensity incident on the eye.

2. HVS. Explain the fundamentals of color vision. Explain the additive and subtractive color mixings.

Objectively, we can distinguish:

Acromatic light (no color): is characterized by the intensity (or amount) of light. It varies from black (no light) to white (very bright light). The intermediate level is gray.

Chromatic light (color) (in the visible spectrum 380-700nm)

that is characterized by 3 quantities:

1. Radiance is a total amount of energy emmited by a source of light (measured in watts (W)).
2. Liminace is a measure of the amount of energy that an observer perceives from the light source.
3. Brightness is a subjective evaluation what is the source of radiation we talk about

Explain the fundamentals of color vision

There are two ways to consider the color vision:

1. we can consider it as a part of a electromagnetic spectrum and from the perspectives of the perception by the human eye and interpretation by the human brain

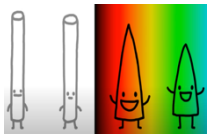
2. To consider it as a part of how the color as such is created by the prime colors (spectrum) or by mixing colors

Color is the way our eyes and brains perceive light . light is a wave and like any type of wave, every particular wave has a wavelength . Technically speaking, the range of human perception is very narrow compared to the complete electromagnetic spectrum .

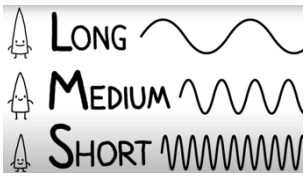
The color we see is based on the wavelength of the light. Also the more there is the brighter the color we see. Most colors are combination of multiple wavelengths.

For example white (as a color) contains all rainbow colors. Newton's prism is used to break up light into its constituent spectral colors because the refractive index depends on frequency

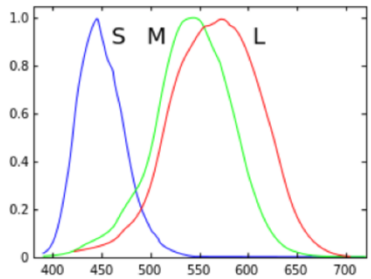
But how do we see these colors? When color enters in our eyes, it hits our retina where there are light-sensitive rod cells and cone cells. Only the cones are used for color vision, so we are just going to focus on them.



There are 3 types of cones and each type is sensitive to different wavelengths in different amounts



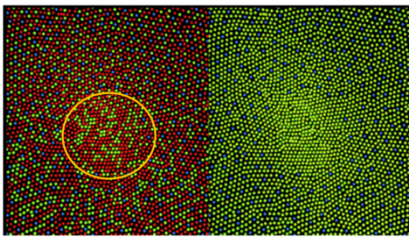
L-cones – sensitive to **high**-wavelength "Red"(not exact)
M-cones – sensitive to **medium**-wavelength "Green"(not exact)
S-cones – sensitive to **short**-wavelength "Blue" (not exact)



But in digital camera We have 2 times more greens receptors rather than red and blue. red and blue have the same proportion.

the largest concentration of receptors for normal person is in the Fovea et not everywhere in retina.

remarque : L- and M-cones are coded in X-chromosome (the cause of color-blondeness) → Genetic determine what we see .
Colorblind people might not have all types of cones or have some issues with them .



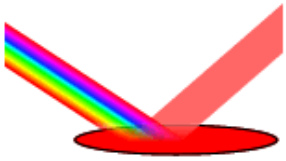
The distribution of cone cells in the fovea of a person with normal color vision (left) and a colorblind (protanopic) retina.
Conclusion:

1. there are no S cones or rods in the fovea.
2. Colorblind people not have L-cones.

Also different animals have different types of cones: Some birds - for 4 or even 5 colors....
Explain the additive and subtractive color mixings:

The additive primary colors are red, green and blue (RGB). Combining one of these additive primary colors with equal amounts of another one results in the additive secondary colors of cyan, magenta and yellow. Combining all three additive primary colors in equal amounts will produce the color white. Remember combing additive colors creates lighter colors, so adding all three primary colors results in a color so "light" it's actually seen as white. Although that may seem strange, if you think of the absence of all light equaling black, it begins to make sense that adding different colors creates white.

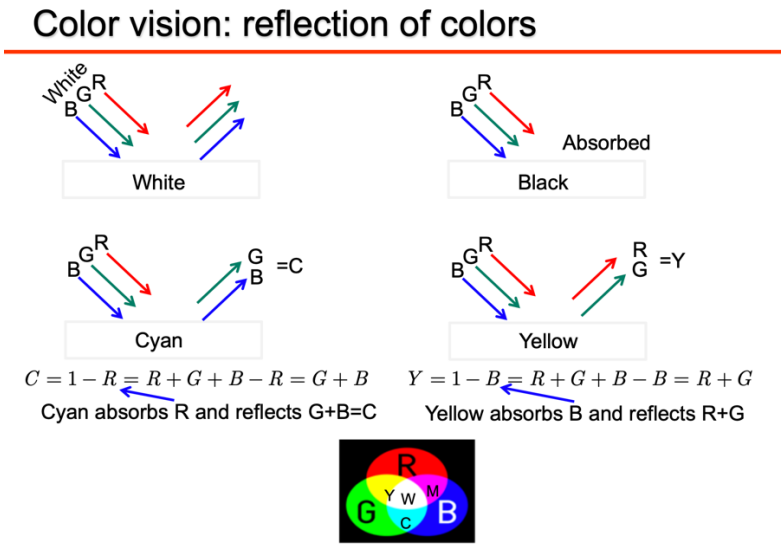
When the source of color is pigment or dye, the result of combining colors is different from when the source of color is light. **Subtractive color** mixing occurs when light is reflected off a surface. For example, a red pigment or filter only appears red because it absorbs (subtracts) all of the light that is not red and only reflects or allows the red light. A green pigment only reflects green light and a green filter only passes green light.

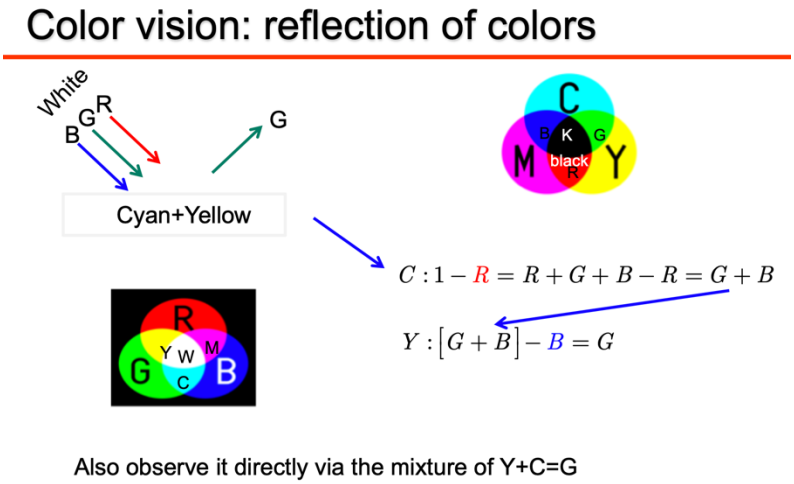


3. HVS. Explain the CMYK model.

Stands for "Cyan Magenta Yellow Black." These are the four basic colors used for printing color images. Unlike RGB (red, green, blue), which is used for creating images on your computer screen, CMYK colors are "subtractive." This means the colors get darker as you blend them together. Since RGB colors are used for light, not pigments, the colors grow brighter as you blend them or increase their intensity.

Exemplify the reflection of colors.





Explain the principles of halftoning.

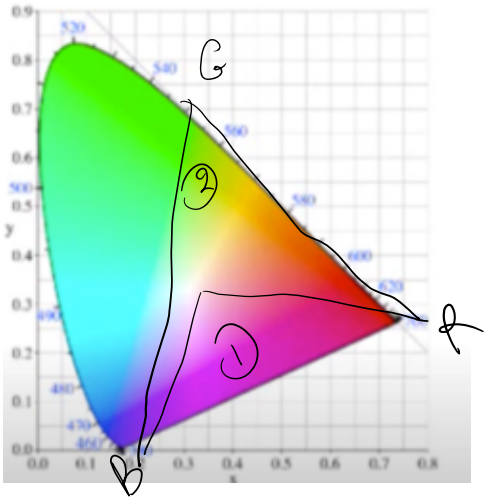
With CMYK printing, halftoning (also called screening) allows for less than full saturation of the primary colors; tiny dots of each primary color are printed in a pattern small enough that humans perceive a solid color. Magenta printed with a 20% halftone, for example, produces a pink color, because the eye perceives the tiny magenta dots on the large white paper as lighter and less saturated than the color of pure magenta ink.[citation needed]

Without halftoning, the three primary process colors could be printed only as solid blocks of color, and therefore could produce only seven colors: the three primaries themselves, plus three secondary colors produced by layering two of the primaries: cyan and yellow produce green, cyan and magenta produce blue, yellow and magenta produce red (these subtractive secondary colors correspond roughly to the additive primary colors), plus layering all three of them resulting in black. With halftoning, a full continuous range of colors can be produced.

Why "K" is used in CMYK model in practice, while it is enough to have CMY to simulate any color? People just for economy reasons uses black inks because black inks is less expensive to produce and **take less time** for printing and **paper is not so wet** and **more saturated** compared by using 3 component.

4. HVS. Explain the chromaticity color diagram.

Before discussing chromaticity color diagram, we need to discuss Hue, saturation, and brightness.
Hue is the dominant wavelength, for example, the redness of red, the greenness of green.
Saturation is how much white is contained in the color, for example, red and royal blue are more saturated than pink and sky blue.
Brightness is an attribute of visual perception in which a source appears to be radiating or reflecting light. it is a subjective attribute/property of an object being observed
The chromaticity diagram shows the full gamut of colors that we can see. Gamut is a reproducible range of colors inside the triangle. Color Gamut is the entire range of colors and tones achievable by an imaging system.



1. Not natural colors because there is no wavelength for these colors as they are obtained by mixing.
2. Achievable gamut of colors

The color gamut of a computer screen is determined by the saturation and brightness of its red, green, and blue pixels. While the color gamut of a printing device is determined by the hue, saturation, and luminance of its cyan, magenta, yellow, and back inks. In addition to the brightness and other characteristics of the paper on which they are printed.

Gamaut of the screen is larger than the gamaut of the printer . printer can reproduce less color

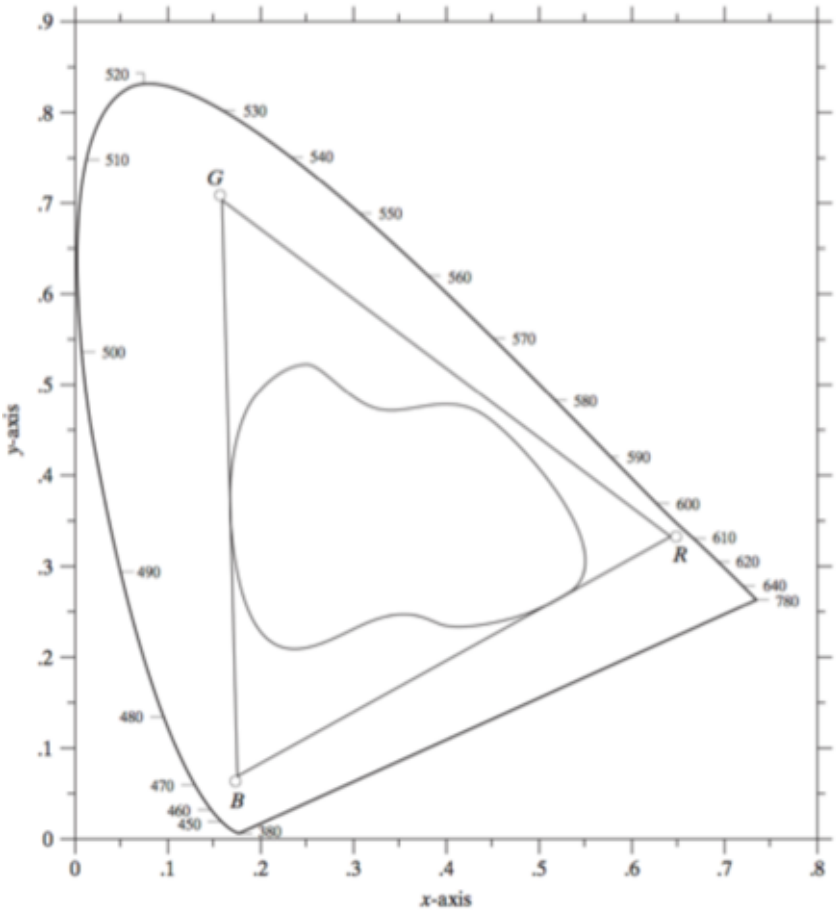


FIGURE 6.6
Typical color gamut of color monitors (triangle) and color printing devices (irregular region).